Five-Month Implementation and Evaluation Plan

Offline Intelligent Support System for Psychological Documentation and Longitudinal Progress Monitoring

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Planned period: 19 October 2026 – 21 March 2027 (22 weeks)

Implementation: Weeks 1–12 **Testing, optimization and pilot:** Weeks 13–22

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1 Objective and delivery strategy

The goal is an **offline, clinician-reviewed system** that turns a consented session recording into timestamped text, an editable clinical-documentation draft, and ultimately a reviewed patient-progress history. Development begins on Linux laptops; later deployment targets the **NVIDIA Jetson Orin Nano Super 8GB**, subject to performance measurements and available hardware.

Immediate commitment: a functional MVP by the end of Week 2. It must process a short fictional audio recording locally, generate a timestamped transcript, populate an editable note draft using attributed excerpts, and save/reopen the session. It is not merely a UI mockup. Automated clinical interpretation and speaker diarization are *not* prerequisites for this first gate.

Scope boundaries: The system supports documentation; it does not autonomously diagnose bipolar disorder, assess clinical safety, prescribe treatment, label emotions, or provide crisis intervention. Clinician-entered observations and a mandatory risk-review step remain central to later note approval. The initial target use cases are bipolar-related longitudinal monitoring and a limited anxiety workflow.

2 System architecture

2.1 Portable software design

Use a **single-machine modular architecture**: local web UI (React/TypeScript), API (FastAPI), background inference worker (Python with pluggable inference runtimes), and local persistence (SQLite plus protected audio files). Only the inference backend and device packaging should require substantial change between x86-64 laptops and ARM64 Jetson hardware.

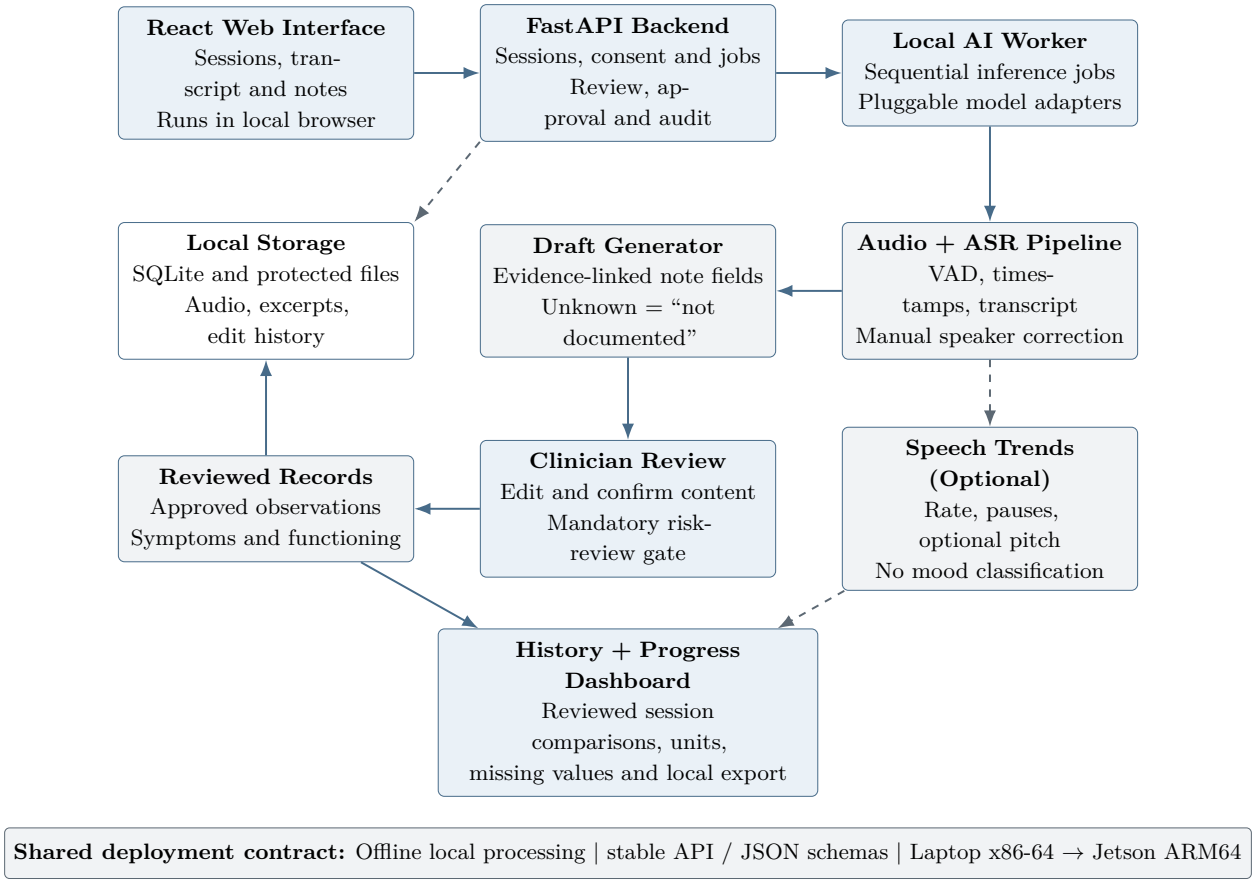


Figure 1: High-level modular architecture. Dashed connections indicate supporting data flow; clinically approved records remain distinct from unreviewed drafts and exploratory measurements.

2.2 Main processing workflow

Consent and audio import → audio preprocessing → timestamped multilingual ASR → speaker correction → source-linked note draft → psychologist editing and mandatory risk review → approved patient history and dashboards.

The MVP uses manually corrected speaker labels and a deterministic, extractive note template. A compact local language model may improve note drafting in later weeks, but every factual statement must be checked against transcript evidence. Heavy AI stages run sequentially to stay within laptop and Jetson memory limits.

3 Milestones and 22-week timeline

Phase A: implementation (Weeks 1–12; approximately three months). Each milestone ends with a working integrated build; major gates are presented to the academic supervisor.

Weeks	Dates	Engineering deliverables	Review gate
1–2	19 Oct–1 Nov	Set contracts, create local UI/API/SQLite, implement short WAV import, offline ASR with timestamps, edit transcript, form source-based draft, save/reopen fictional session.	**M1: Functional MVP** Professor demo, W2

Weeks	Dates	Engineering deliverables	Review gate
3–4	2–15 Nov	Add consent states, error handling, audio checks, source excerpt links, manual speaker labels, note-template configuration and repeatable local installation. Begin recording fictional scenarios.	Internal integration, W4
5–6	16–29 Nov	Add optional local LLM-assisted draft generation (strict JSON and source checks), clinician note corrections, objective fields, review status and blocked approval until risk review.	**M2: Clinical workflow** Professor + consultant, W6
7–9	30 Nov–20 Dec	Implement patient history, reviewed symptoms/functioning, interventions, missing-data rules, charts and timeline. Test speaker attribution baseline on two-person recordings.	**M3: Monitoring beta** Professor, W9
10–12	21 Dec–10 Jan	Complete local app flow, exploratory speech-rate/pause panel, basic Urdu–English evaluation, audit and privacy controls, unit/integration tests, packaging, and implementation code freeze.	**M4: Feature complete** Professor, W12

Phase B: evaluation, performance, hardware and industry pilot (Weeks 13–22). No major feature expansion after the Week 12 freeze unless needed to resolve a critical defect.

Weeks	Dates	Evaluation / deployment deliverables	Review gate
13–16	11 Jan–7 Feb	Held-out fictional scenarios; English, Urdu and code-switch ASR testing; speaker errors; invented/omitted note facts; reviewer effort and risk-workflow tests; fix high-severity failures.	**M5: Quality review** Professor + consultant, W16
17–19	8–28 Feb	Benchmark 45–60 min sessions, peak RAM/VRAM, inference latency and offline behavior. Select quantized models; decide whether to port to Jetson Orin Nano Super. Try ARM64 deployment only if feasible.	**M6: Jetson go/no-go** Professor, W19
20–22	1–21 Mar	Conduct supervised industry-workflow pilot using fictional cases, gather usability feedback, repair defects, produce reproducible demo, deployment guide, metrics and final report.	**M7: Final sign-off** Professor, W22

4 Week 2 MVP: exact definition of done

1. A user selects a fictional patient/session and imports a **2–5 minute WAV recording** through the local application.
2. The worker runs a **locally downloaded ASR model** without an internet connection and returns timestamped transcript segments. A correct transcript is *not* assumed; text can be edited.
3. A **structured draft** is automatically populated with selected, time-linked transcript excerpts (at minimum, subjective statements). Unavailable clinical fields display *Not documented*. No diagnosis is generated.
4. The user can edit the transcript and note, **save both in SQLite**, close the app and reopen the same session with its edits preserved.
5. A failed recording or processing job produces a visible error rather than a blank screen; the app can be installed and demonstrated on a team laptop.
6. The team records **actual** processing time and known defects, then demonstrates the workflow to the professor using only fictional or consented role-play audio.

MVP exclusions: reliable automatic diarization, 60-minute sessions, autonomous clinical summaries, polished charts, Jetson deployment and claims of clinical validity. The Week 2 build remains in *draft* state; its user interface must not imply that clinical sign-off or risk screening already works.

5 Team responsibilities and dependencies

Owner	Implementation (Weeks 1–12)	Evaluation / pilot (Weeks 13–22)
Anas AI + integration	Audio/ASR adapters, timestamp schema, transcript evidence, local note generation, speaker pipeline and exploratory speech features; lead integration and model selection.	WER/CER and speaker benchmarks, note-grounding errors, GPU/RAM profiling, quantization and Jetson compatibility; model documentation.
Talha (Taha) API + security	FastAPI endpoints, schema migrations, local job status, patient/session DB, consent and approval rules, source references, auditing and storage protections.	Performance logging, offline/network tests, backup/restore and deletion checks, deployment packaging, security and API failure tests.
Ayyan (Ayaan) UI + evaluation	React/TypeScript UI, audio import, transcript/note editors, risk-review screens, timelines/charts, fictional cases and usability flows.	Annotated evaluation cases, error review, usability timing, consultant feedback, documentation, final demonstration and report figures.

Team integration rule: Agree the audio, transcript, evidence and note JSON schemas in Week 1. Until the real worker is ready, use the *same JSON structure* as a mock response. Merge reviewed pull requests at least twice weekly; run a joint working demo each Friday. Each module needs tests, sample inputs, and a clear hand-off contract.

6 Technology, data and deployment decisions

Component	Recommendation
Frontend	React + TypeScript + Vite; Recharts for longitudinal visualizations.
Backend	Python + FastAPI + Pydantic + SQLAlchemy/Alembic; a separate local processing worker.
Local persistence	SQLite for clinical structure and audit; protected audio directory. SQLite itself is not encrypted: use an encrypted-volume or SQLCipher-backed approach before any approved sensitive use.
Speech recognition	Start with multilingual whisper.cpp (small/base benchmark); compare faster-whisper on the laptop. Runtime/backends are hardware-specific.
Note generation	Start with an extractive, auditable template; evaluate a quantized GGUF local model through llama.cpp only after an evidence-linked baseline works.
Speech features	NumPy and audio-timing utilities for speaking rate and pause duration; pitch variation only if extraction is reliable.
Test datasets	Team-authored fictional role-play sessions (priority), FLEURS, licensed Common Voice Urdu samples, and AMI for overlapping multi-speaker audio. Check current dataset licenses before use.
Device portability	Keep model paths, acceleration options and worker implementations configurable; benchmark x86-64 Linux first, ARM64/JetPack later. Avoid assuming CUDA binaries transfer directly.

Data collection starts in Week 1, not after implementation. Create short, fictional cases with manual transcripts and an expected note outline. By Week 12, aim for approximately 15–20 consented fictional recordings across English, Urdu and mixed-language settings, with multiple sessions per fictional client; this is a planning target, not a guarantee of statistical power. Reserve separate cases/speakers for held-out evaluation.

7 Evaluation, safety and industry-workflow trial

Track (a) transcription error rates with language-specific normalization, (b) speaker attribution mistakes, (c) evidence-linked note omissions and unsupported claims, (d) clinician correction time versus manual drafting, (e) consistency of units and missing values on progress charts, (f) correctness of risk-review approval rules, and (g) time, peak RAM/VRAM and recording reliability on realistic 45–60 minute inputs. Agree numerical acceptance thresholds with the professor after the baseline pilot; do not invent benchmark results.

Industry implementation in Weeks 20–22 means a supervised workflow demonstration with a psychologist or clinic partner using *fictional* records, followed by feedback and bug fixes. Real clinical recordings require separate institutional ethics approval, explicit consent, security and data-governance arrangements; they are *not* assumed within this five-month plan.

Jetson decision rule: Proceed only if a 45–60 minute benchmark, memory headroom, acceptable note quality, ARM64 runtime compatibility and full hardware budget support deployment. If the Jetson is unavailable or does not meet the workload, the evaluated laptop build remains the primary deliverable with a documented feasibility report.

8 Supervisor review procedure and final deliverables

At each M1–M7 review, bring: **(1)** a live demonstration, **(2)** completed versus planned tasks, **(3)** measured results and known errors, **(4)** key decisions requested, and **(5)** the next sprint backlog. Record the professor's action items with an owner and due date.

Final handover (Week 22): offline installer and source code, versioned models/configurations, patient/session and approval workflow, progress-dashboard demonstration, evaluated synthetic dataset manifest, ASR and note-quality results, laptop performance report, optional Jetson build/benchmark, security limitations, industry pilot feedback, user guide and FYP report.